Problem Sheet for Week 3

1. Write a program to perform LU decomposition followed by forward and backward substitution to solve $\mathbf{A} \cdot \vec{x} = \vec{b}$ for $\vec{x}$. Use this to also invert the matrix $\mathbf{A}$.

   Try your code with the 5×5 matrix $\mathbf{A}$ and vector $\vec{b}$

   $$\mathbf{A} = \frac{1}{6}\begin{pmatrix} 5 & 4 & 3 & 2 & 1 \\ 4 & 8 & 6 & 4 & 2 \\ 3 & 6 & 9 & 6 & 3 \\ 2 & 4 & 6 & 8 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \tag{1}$$

   You can get a measure of the accuracy of your result by calculating the residuals, either the vector $\vec{r} = \mathbf{A} \cdot \vec{x} - \vec{b}$ when solving for $\vec{x}$ or the matrix $\mathbf{R} = \mathbf{A} \cdot \mathbf{A}^{-1} - \mathbf{I}$ when solving for $\mathbf{A}$. In the absence of errors, these would have zero in every element. Summarise the accuracy using something such as the average residual magnitude $\sum_i |r_i|/5$ or $\sum_{ij} |R_{ij}|/25$, respectively.

   Finally, code up the pattern of the numbers in $\mathbf{A}$ so you can make an arbitrarily big $N \times N$ matrix. Note, the multiplicative fraction at the front is $1/(N+1)$. How big can N be before your code to solve for $\vec{x}$ starts taking a ridiculously long time to run? How does the average residual depend on the matrix size?

2. Implement the method of powers to find the eigenvalue with the largest modulus of the 5×5 matrix $\mathbf{G}$

   $$\mathbf{G} = \begin{pmatrix} 0 & -1/2 & 0 & 0 & 0 \\ -1/2 & 0 & -1/2 & 0 & 0 \\ 0 & -1/2 & 0 & -1/2 & 0 \\ 0 & 0 & -1/2 & 0 & -1/2 \\ 0 & 0 & 0 & -1/2 & 0 \end{pmatrix} \tag{2}$$

   You will need to decide on a stopping criterion based on the change of the eigenvalue or the average change to the eigenvector. You should also catch non-convergence in case of problems.

   How does the eigenvalue vary with the size of the matrix, when keeping the pattern of the numbers the same?

   Try this with $-1/3$ rather than $-1/2$ along both off-diagonal lines.

   [**Hint**: If your first implementation of the method does not work, what special structure does the matrix have that might prevent the technique from working? How might you adjust the method?]

3. Implement the iterative Jacobi methods to solve for $\vec{x}$ and to find the inverse of $\mathbf{A}$. Run it using

   $$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 5 \end{pmatrix} \tag{3}$$

   You will need to choose a stopping condition for the iteration loop. For a small matrix such as the above, it will not be excessive to calculate the residual vector or matrix for every iteration step and use the sum of the magnitudes of the elements as a measure of the convergence.

   You will also need to prevent the program getting stuck in an infinite loop if the iteration does not converge. Can you modify $\mathbf{A}$ to purposefully make the method not converge so as to check this?

   This matrix is *not* strictly diagonally dominant but the Jacobi method does actually converge. However, it can take many iterations to do so and the number of iterations required gets very large as the matrix gets bigger. See how the number of iterations grows with the matrix size. Compare with the same size matrix but with 3's along the diagonal rather than 2's. The 3's make the Jacobi approximation better and it should converge much faster.

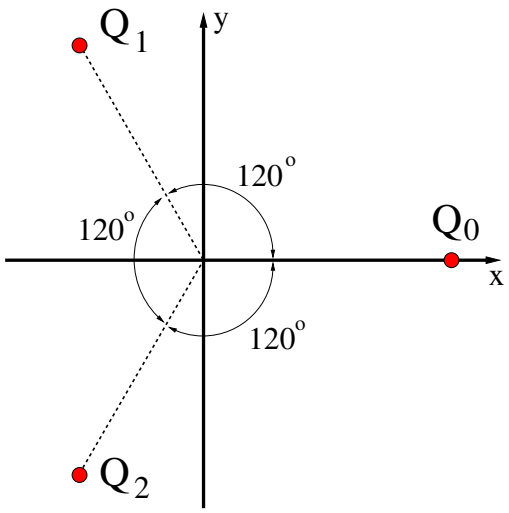
   Try to link your results to your answers for Qu 2.

4. Finding the position between several point charges at which there is no electric field.

   The vector form of the Coulomb law for the electric field from point charges Q_i at positions $\vec{r}_i$ is

$$\vec{E}(\vec{r}) = \sum_i \frac{Q_i(\vec{r} - \vec{r}_i)}{4\pi\epsilon_0|\vec{r} - \vec{r}_i|^3} \tag{4}$$

   If all the charges have the same sign, there can be a position within the region of the charges where the vector contributions to the electric field from all the charges cancel out and so a test charge placed there would not move.



   Consider the three positive charges in the $z = 0$ plane as shown in the above figure, each a distance 1 m from the origin and at plane polar angles of 0°, 120° and 240° degrees. The charge at 120° is half that of the other two, i.e. $Q_0 = 2Q_1 = Q_2$.

   Since it is clear by symmetry that the $\vec{E} = 0$ point must also be in the $z = 0$ plane, the calculation can be done in only two dimensions, x and y. There is no natural length scale for this problem but as the charges are of O(1 m) apart, then metres are a suitable unit. We can scale the electric field using

$$\vec{E}'(\vec{r}) = \frac{4\pi\epsilon_0}{Q_0}\vec{E}(\vec{r}) = \sum_i \left(\frac{Q_i}{Q_0}\right) \frac{(\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3} \tag{5}$$

   where $Q_0/Q_0 = Q_2/Q_0 = 1$ and $Q_1/Q_0 = 1/2$, so all terms will be O(1).

   Use a generalised Newton-Raphson method for two variables, with either analytic or numerical estimates for the derivatives in the Jacobian matrix, to find the position at which $\vec{E}' = 0$. You will need to pick a sensible starting guess for x and y; try to think of arguments using symmetries or slightly modified geometries which can help with this.

Questions 5 and 6 are for discussion in the Wednesday practical sessions (Starting Week 4).

5. Taylor series:

   Pretend your computer cannot do division. We want to calculate $1/x$, so we will use a numerical method.

   Writing $x = 1 + g$, the first few terms of the Taylor expansion of the function $f(x) = 1/x$ around the point $x = 1$ are

   $$f(1+g) = \frac{1}{1+g} \approx 1 - g + g^2 - g^3 + \ldots \tag{6}$$

   The "zeroth-order" approximation for small g is just the first term, so $f(1+g)^{(0)} = 1$. The first-order is $f(1+g)^{(1)} = 1 - g = 1 - g\, f(1+g)^{(0)}$ and the second-order is $f(1+g)^{(2)} = 1 - g + g^2 = 1 - g\, f(1+g)^{(1)}$. The same relation to the previous approximation holds for all higher terms as well, so we can estimate $1/x$ by *iterating* our solution according to

   $$f(1+g)^{(i+1)} = 1 - g\, f(1+g)^{(i)} \tag{7}$$

   starting from $f(1+g)^{(0)} = 1$.

   a) Write a function to implement this iteration for a given value of g. You will need to decide when to stop taking further iterations by picking a *stopping condition*. You could pick a fixed number of steps, which is equivalent to a fixed order of the Taylor series. However, the error on the estimate arising from the remainder term will then depend strongly on the value of g. It is more typical to stop the iteration when the magnitude of the relative change to the result becomes smaller than a defined value; you could (arbitrarily) use 10^{-6} for this exercise. Pick a few values of g (choose both positive and negative values) and compare your iteration result to the true value of $1/(1+g)$. Check the error, i.e. the difference from the true value, is approximately consistent with your chosen convergence criterion value.

   b) You should find the estimate converges quickly for $|g| \ll 1$. However, as $|g| \to 1$, it will take more and more iterations before it converges. When $|g|$ gets very close to 1, you may find it takes too long to converge and you may have to ensure you stop the iteration differently in this situation. Check these statements hold for your implementation.

   c) Step through a range of values of $|g| < 1$ and make plots of the value of the solution, the relative (fractional) error on the solution, and the number of iterations needed, each as a function of x.

   d) You should have found this method will not converge for $x \geq 2$. However, can you spot a way to use your code to evaluate e.g. $1/2.1$ without needing to rewrite it? Hint: you know $1/2 = 0.5$ without having to use division on your computer.

6. **Former demonstrator Mike Heyns' question:** Interpolating the Earth's geomagnetic field using an equivalent current system.

   One of the challenges in geophysical and space weather modelling is estimating the horizontal components of Earth's geomagnetic field given sparse geomagnetic field measurements. To get around this, we can estimate an equivalent near-Earth current system and integrate the result to find the geomagnetic field at an arbitrary point. More details on this technique can be found in https://doi.org/10.1007/978-3-030-26732-2_2.

   Using the southern African context (see map below), we will use real data from three geomagnetic observatories in South Africa and Namibia, known as HER, HBK and TSU, to estimate the geomagnetic field at a fourth observatory KMH and compare with the measured values there.

   Data files of the surface magnetic field measurements taken every minute by these different observatories for 48 hours during a geomagnetic storm, called the "St. Patrick's Day Storm of 2015", are available from Canvas. The data fields are time [UTC] and geomagnetic field [nT] in the x (northerly) and y (easterly) directions.

   To set up the problem, let us assume six infinite line currents parallel to the Earth's surface in space above the three geomagnetic observatories, i.e. separate currents in the north-south (x) and east-west

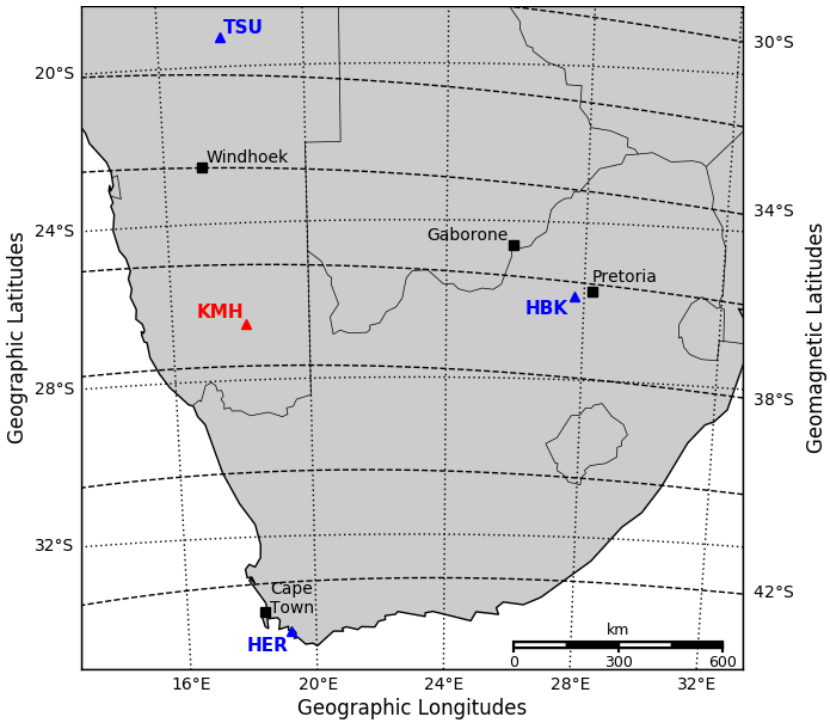


Figure 2: Geomagnetic observatories in southern Africa.

(y) directions above each. Note the right-handed coordinate system with z downwards. Taking both horizontal components into account, the matrix equation at any specific time would be

$$\begin{bmatrix} T_{1,1} & \cdots & T_{1,2n} \\ \vdots & \ddots & \vdots \\ T_{2m,1} & \cdots & T_{2m,2n} \end{bmatrix} \begin{bmatrix} I_{y:1} \\ \vdots \\ I_{y:n} \\ I_{x:1} \\ \vdots \\ I_{x:n} \end{bmatrix} = \begin{bmatrix} B_{x:1} \\ \vdots \\ B_{x:m} \\ B_{y:1} \\ \vdots \\ B_{y:m} \end{bmatrix} \qquad \text{i.e.} \qquad \mathbf{T} \cdot \vec{I} = \vec{B}\,, \tag{8}$$

where $\vec{B}_{c:i}$ is the geomagnetic field component along coordinate $c = x, y$ on the ground at position $i = 1...m$; vector $\vec{I}_{c:j}$ is the orthogonal current amplitude component at position $j = 1...n$, and $\mathbf{T}$ is the transfer matrix relating the two. In this case, we assume the number of current elements n to match the number of magnetometers m, i.e. $\mathbf{T}$ is a square matrix. Making use of the Biot-Savart relation for an infinite line current (see https://doi.org/10.1046/j.1365-246x.1998.00388.x), the resulting circulating magnetic field a perpendicular distance R away has a magnitude $\mu_0 I/2\pi R$. Including resolving the magnetic field into components, we find $\mathbf{T}$ is simply a spatial mapping, i.e.,

$$\begin{aligned} T_{i,j} &= \frac{\mu_0}{2\pi} \frac{h}{\left[h^2 + (x_i - x_j)^2\right]} && \text{for } B_{x:i} \text{ resulting from } I_{y:j} \\ &= -\frac{\mu_0}{2\pi} \frac{h}{\left[h^2 + (y_i - y_j)^2\right]} && \text{for } B_{y:i} \text{ resulting from } I_{x:j} \end{aligned} \tag{9}$$

where h is the height of the equivalent current system (for simplicity assume $z = 0$ at the current system height and $z = h$ at the surface of Earth).

(a) As a preliminary step, factorise the elements of $\mathbf{T}$ into an overall constant and a dimensionless value that is different for each element. Consider the size of your values and whether further scaling is appropriate.

Since the matrix equation is valid for the input data at all times (since the spatial mapping between currents and magnetometers is invariant in time), it pays to compute the inverse of $\mathbf{T}$ once and then

apply it for all times. If you are feeling up to it, try to compute the individual elements in $\mathbf{T}$. Assume $h = 1000\,\text{km}$ and the table below gives other details that are needed. Hint: At mid-latitudes the dominant driving current systems lie in the magnetosphere, a few Earth radii away. Given the separation distances between magnetometers we can assume a flat geometry is a decent approximation. Hence, a quick and dirty arc length computation is sufficient and should give something similar to the last two columns.

Station	Subscript	Latitude	Longitude	Easterly (y) Position	Northerly (x) Position
HER	1	-34.43°	19.23°	≈150 km	0 km
HBK	2	-25.88°	27.71°	≈1000 km	≈950 km
TSU	3	-19.22°	17.7°	0 km	≈1700 km
KMH	-	-26.54°	18.11°	≈40 km	≈900 km

Table 2: Details for spatial mapping as defined using an arbitrary grid. The "Subscript" column relates to the order of mapping in $\mathbf{T}$ given below (i.e. the n and m indices in the matrix equation).

(b) Write a program to use row manipulations which can perform **Gauss-Jordan elimination** to find the inverse $\mathbf{T}^{-1}$. If you haven't computed the elements of $\mathbf{T}$ for mapping HER, HBK and TSU to the equivalent current system, use the following:

$$\mathbf{T} = \begin{bmatrix} 1 & 0.526 & 0.257 & 0 & 0 & 0 \\ 0.526 & 1 & 0.64 & 0 & 0 & 0 \\ 0.257 & 0.64 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -0.581 & -0.978 \\ 0 & 0 & 0 & -0.581 & -1 & -0.5 \\ 0 & 0 & 0 & -0.978 & -0.5 & -1 \end{bmatrix}. \tag{10}$$

Note that the form of $\mathbf{T}$ given excludes a constant $(\mu_0/2\pi h)$ and uses a normalisation of (r/h). Also ensure that you have your magnetic field vector in the right order, i.e. HER, HBK and TSU as defined by the subscripts in the matrix equation and the table above.

(c) Now use the inverse $\mathbf{T}^{-1}$ to estimate the equivalent current system $\vec{I}$ from the measured geomagnetic field $\vec{B}$, i.e. $\vec{I} = \mathbf{T}^{-1} \cdot \vec{B}$. Once you have $\vec{I}$, simply apply the same mapping approach to KMH, i.e. $\vec{B}' = \mathbf{T}' \cdot \vec{I}$, where $\mathbf{T}'$ is the matrix for mapping at KMH. If you haven't computed the mapping, use

$$T' = \begin{bmatrix} 0.552 & 0.998 & 0.61 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.988 & -0.52 & -0.998 \end{bmatrix}. \tag{11}$$

Apply this inverse to all timesteps, possibly optimizing though vectorization. You can then plot the calculated values and measured data at KMH to compare them; you should be able to produce something similar to the figure below.

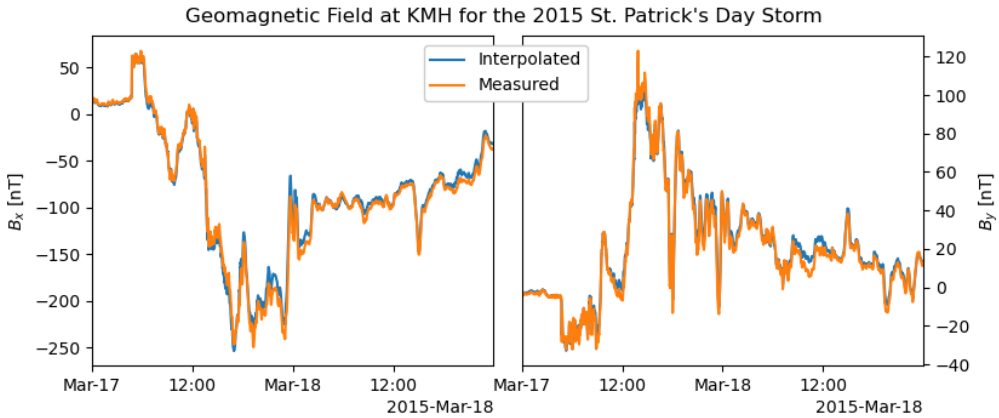


Figure 3: Comparison of interpolated and measured geomagnetic fields.

This figure was made using $h = 1000$ km, which is around the separation between magnetic observatories used. How does the interpolation change for a change in h? In mid-latitude regions magnetospheric currents situated at multiple Earth radii drive the geomagnetic disturbances seen at the surface. Looking at the interpolation performance and vector $\vec{I}$, can we say anything about the nature of these currents?

(d) There are often cases when we do not want to reuse an inverse, or perhaps have too large a matrix to invert. We can set up this calculation to have a very large matrix by including data from multiple times in a single matrix equation, i.e.,

$$\begin{bmatrix} [T]_{t_0} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & [T]_{t_f} \end{bmatrix} \begin{bmatrix} [I]_{t_0} \\ \vdots \\ [I]_{t_f} \end{bmatrix} = \begin{bmatrix} [B]_{t_0} \\ \vdots \\ [B]_{t_f} \end{bmatrix}, \tag{12}$$

where $[T]$, $[I]$ and $[B]$ are repeated matrices/vectors in the form of the original matrix equation for each time, ranging from $t = t_0$ through to the final desired time $t = t_f$. For increasing t_f, even this sparse matrix becomes increasingly costly to invert.

Verify whether an iterative is applicable, and then using the first 1000 timesteps implement a **Jacobi or Gauss-Seidel method** with an appropriate stopping condition to compute a single vector $\vec{I}$ for all timesteps. Do both methods converge? As before, apply this to interpolate the geomagnetic field at KMH. Compare to both measured data and results from the previous method.

(e) (**Bonus**) What would happen for an underdetermined problem when there are less magnetometers than current elements? Using your existing code, can you include additional flatness or smoothness constraints and possibly produce a more physical equivalent current system? Hint: Can you use transposes to frame the problem in terms of square matrices?

Data Acknowledgement:
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